

Take some of the most important thinkers of our times. Sir Isaac Newton, as we all know, discovered gravity when an apple dropped on his head. When put so simply with our modern understanding of gravity, it seems like an everyday occurrence and only natural that he thought of gravity from that point. But try and put yourself in his shoes, in a time when the concept of gravity did not exist. Without using the creative muscles of his brain, Newton could never have thought outside the box of then-prevalent scientific reasoning to come up with a new theory about what made the apple fall from the tree.

For a modern example, let's look at the late Steve Jobs. Love him or hate him, you can't deny the fact that he spearheaded several projects that redefined technology, like the iPod. And let's face it, it wasn't that Jobs revolutionised technology by inventing some new metal or discovered a new process. He used his creativity to solve the problem of using existing technology in the field of portable music. It was a new way of thinking about an old problem – and such creativity is crucial to problem-solving.

So any modern artificial intelligence system could not be said to be true AI without the ability to think creatively. It would need to come up with solutions that have eluded the thought process of its programmers – and perhaps even escaped their subconscious.

The problem with “programming” creativity

Now that we have some sort of a loose definition for creativity, it's important to understand the basic problem that has dogged everyone in the field of artificial intelligence: if we accept that creativity is the ability to come up with a new way of thinking, how do we program it into a machine? The very nature of programming goes against this.

Programming, in essence, is laying down a set of instructions to give to a machine, which it duly carries out. Now, these sets of instructions are laid down by a human. So conceptually, the human has thought of these instructions. But how does a human instruct a machine to think of something he has never thought of? You can program the machine to think, of course, but how do you program a machine to think about something that is outside the realm of thought process itself?

More importantly, programs are task-oriented. An intelligent agent is fed a series of instructions that make it calculate and reason with a very specific objective in its digital mind. With creative thinking, what is the objective?